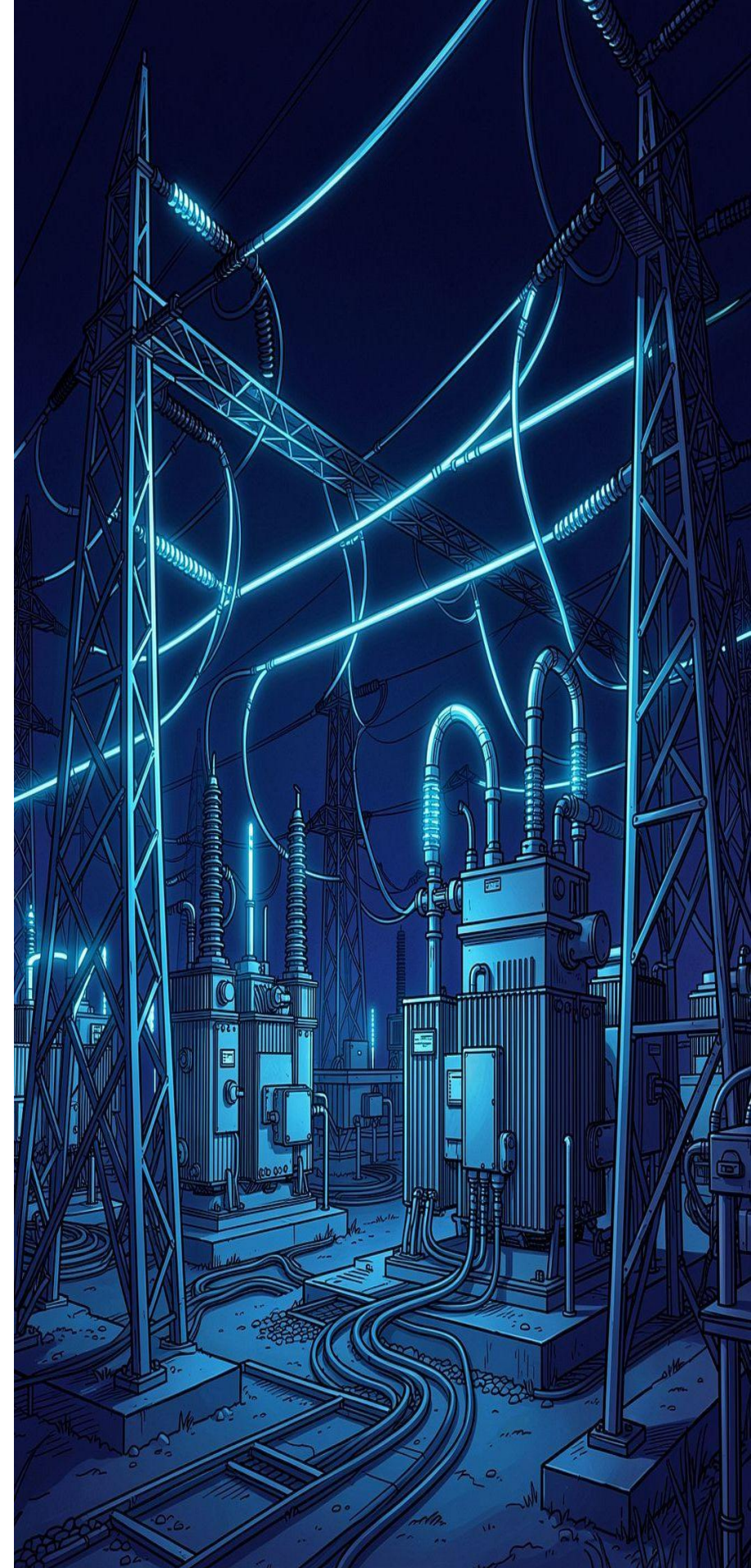


# **GridGuard AI**

## **Predict failures. Prioritize maintenance. Protect the grid.**

Team VoltVision • IBM Bob AI Innovation Hackathon • Track: AI



PROBLEM STATEMENT — U1

# The Grid Is Failing Blind

## Reactive, Not Preventive

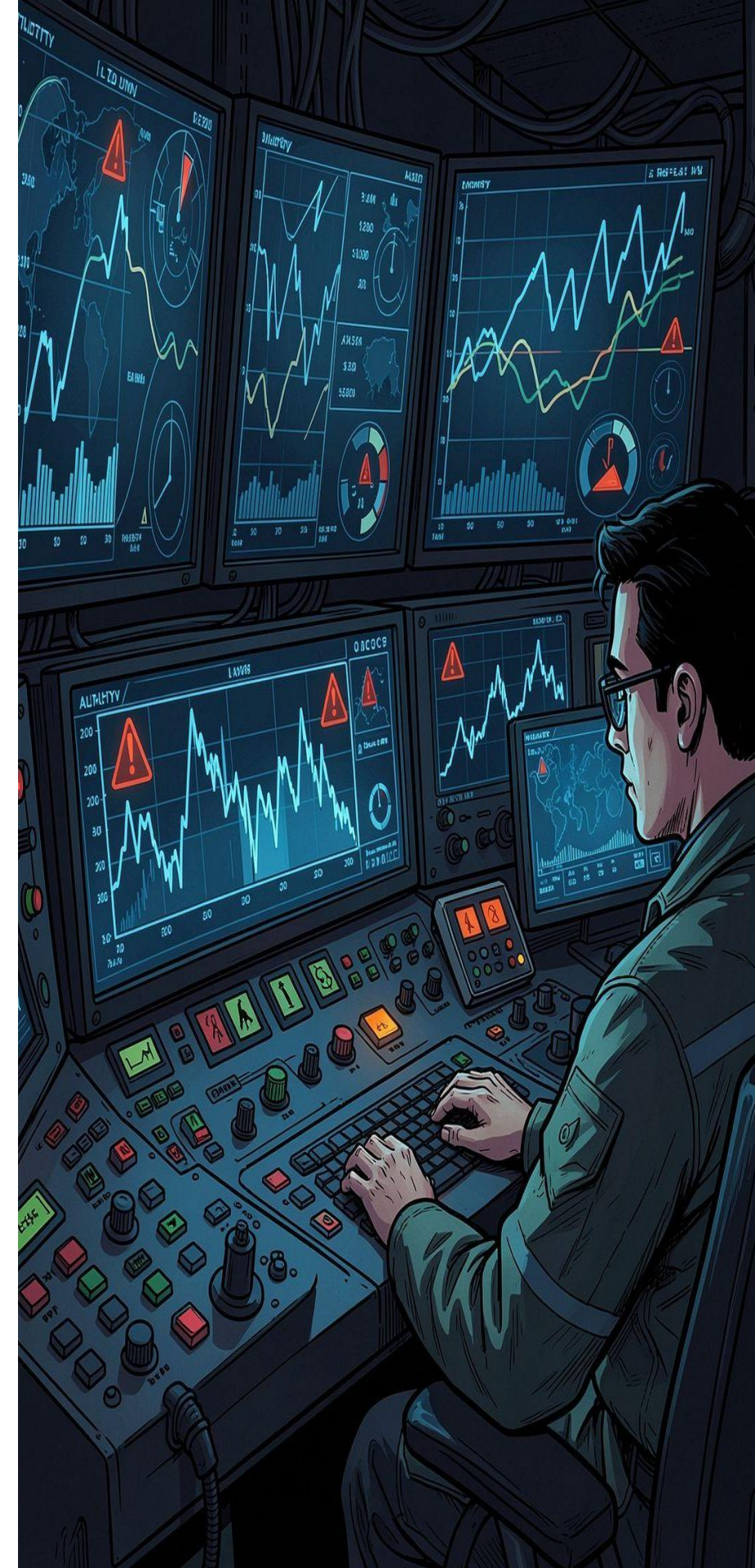
Outages happen before action is taken.

## Fragmented Data

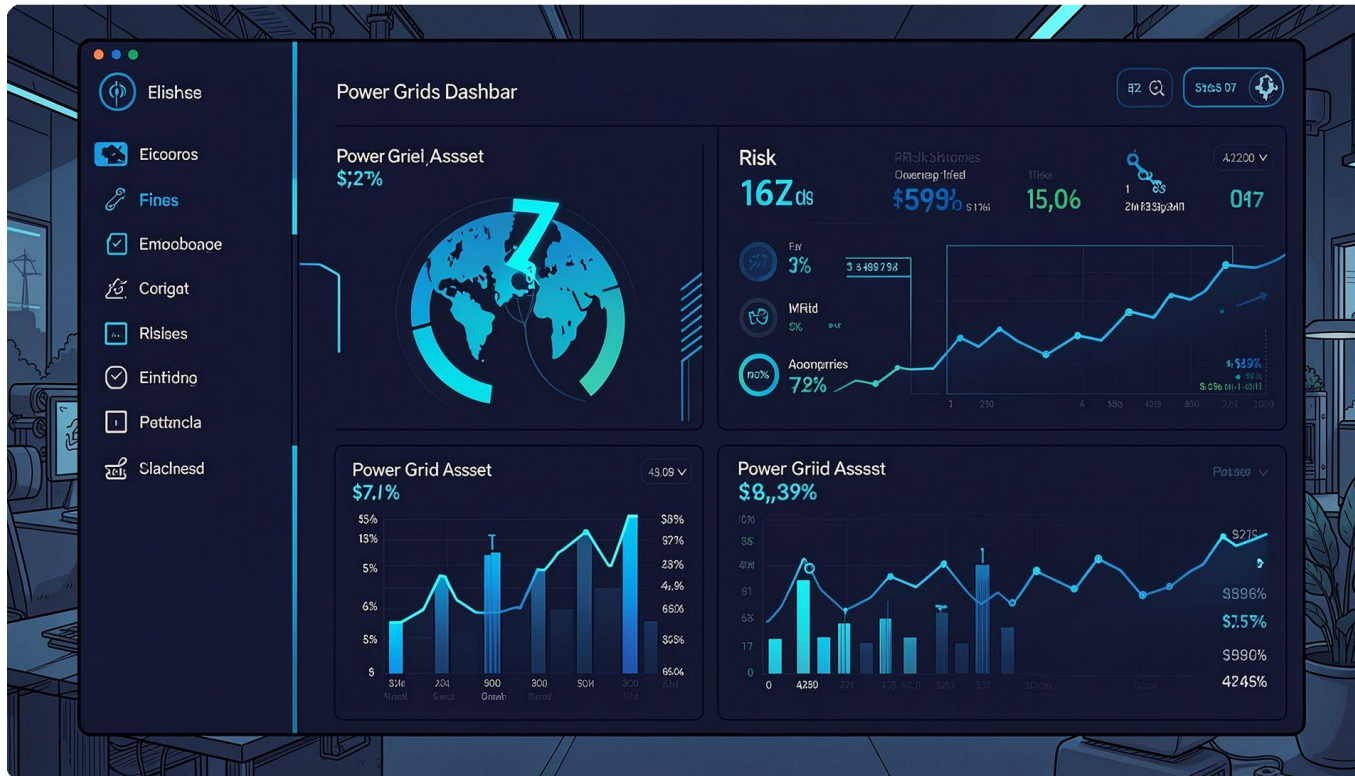
Sensors, weather, and incident records are siloed.

## No Prioritization Logic

Which asset fails first? Where should crews go?



# Introducing GridGuard AI



GridGuard AI fuses four critical data streams into a single, explainable risk assessment.

Output: **Risk score** · **Priority ranking** · **Actionable recommendations**

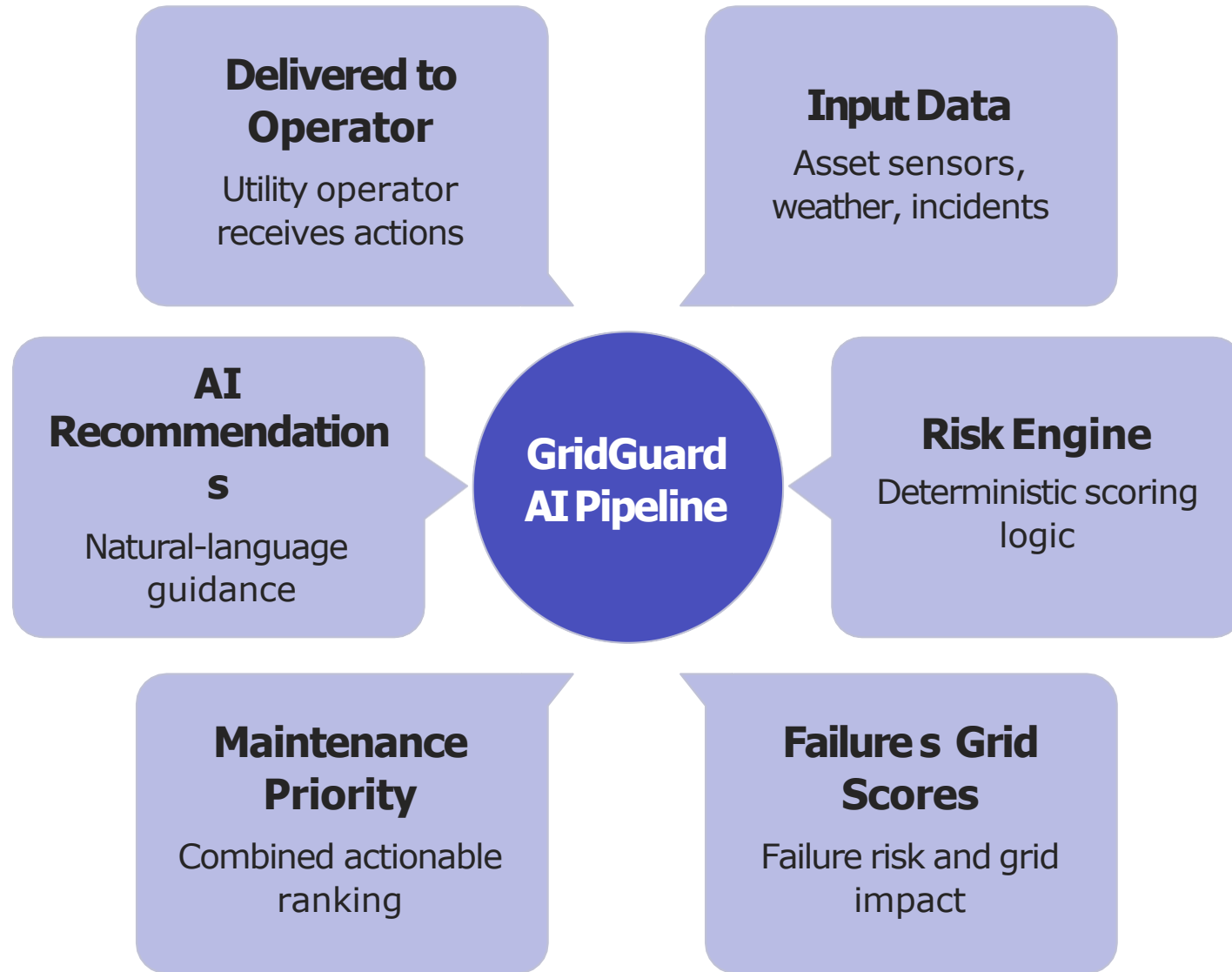
→ **Asset Health Sensors**

→ **Weather Risk Data**

→ **Historical Incident Records**

→ **Grid Impact Analysis**

# How GridGuard AI Works



Numerical risk scores use transparent deterministic logic. AI generates natural-language explanations and maintenance recommendations.

# The Maintenance Priority Formula

$$\text{Maintenance Priority} = \text{Failure Risk} \times \text{Grid Impact}$$

## Failure Risk

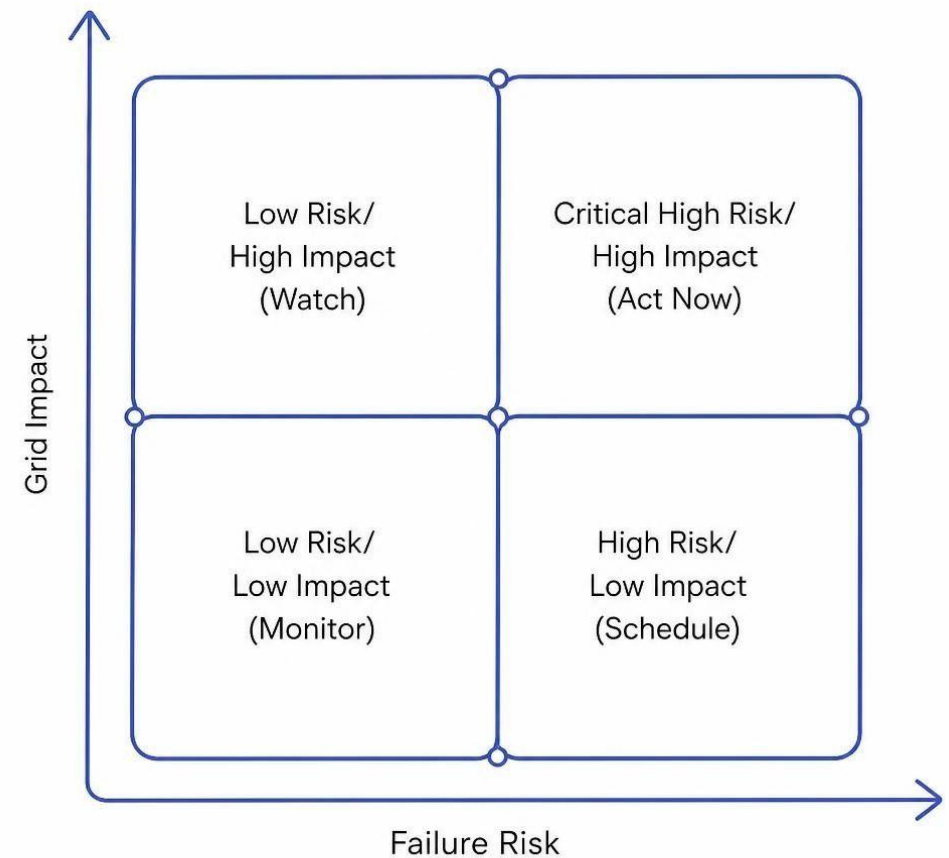
How likely is this asset to fail?

## Grid Impact

How serious would the failure be?

## Maintenance Priority

What should the team act on first?



# Transformer TR-102 — Critical Alert

**Temperature**  
91°C

**Vibration**  
High

**Oil Quality**  
Poor

**Weather Risk**  
Severe

**Historical Incidents**  
3

87%

Failure Risk

94%

Grid Impact

81.8

Maintenance Priority

 TR-102 requires immediate inspection — highest-priority asset in the current risk window.

# From Score to Action

## AI Recommendation — TR-102

**"Inspect TR-102 within 24 hours."**

Contributing factors:

- ♦ High temperature · Abnormal vibration
- ♦ Poor oil quality · Severe weather
- ♦ Historical failure pattern

Actions: Dispatch crew · Inspect cooling · Check oil · Pre-position equipment

## IBM Integration

### IBM Bob IDE

AI development partner for planning, coding, and testing

### watsonx.ai

Planned runtime layer for NL explanations and recommendations

### Local Fallback

App stays functional without external AI credentials

IMPACT & VISION

# See the Risk Before the Outage

GridGuard AI shifts utilities from **reactive** to **risk-based preventive** decision-making.



**Real-Time IoT Streams**



**Live Weather APIs**



**SCADA / OMS Integration**



**Automated Crew Scheduling**

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GridGuard AI — VoltVision

Made with **GAMMA**